PropertyScope NSW

Release 2 / Technical report

41026 Advanced Software Development — Assessment 3

Submission	Group 20
Team	Matthew Shelton, Burhan Naeem, James Huang, Michael White, Derek Song
Due	25 October 2026, 11:59 pm Sydney time
Repository	MattShelton04/41026ASDProject
Showcase video	Pending
Showcase	23 October 2026
Commit reference	main

TO COMPLETE / GROUP

Replace "Pending" with the published showcase video link, and use the same URL in RELEASE2_METADATA and Section 14

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1 Project overview

PropertyScope NSW helps buyers research NSW property. Each student owns a containerised frontend, backend/API and database slice; Matthew also leads the shared platform. Release 2 keeps the Release 0–1 AI-mode, agentic loop, MCP and RAG servers, and adds:

- a host Multi-Agent Server (Planner → Worker → Reviewer → human review), called through a feature's backend;
- three review modes: multi-agent workflow, testing report and cloud deployment;
- pre-commit security scans, and a `student-N.yml` per student for post-commit endpoint tests;
- Azure deployment with the AI tier disabled by default.

Student	Feature	Release 2 responsibility
1 Matthew Shelton (24763373)	F1 Data platform and property discovery	Shared platform (Multi-Agent Server, review modes, security scans, test kit, Azure); F1 workflow and endpoint tests
2 Burhan Naeem (24764134)	F2 Property sales and market cases	F2 workflow, endpoint tests and evidence
3 James Huang (24970865)	F3 Suburb, crime and liveability analytics	F3 workflow, endpoint tests and evidence
4 Michael White (24846267)	F4 Site, planning and building due diligence	F4 workflow, endpoint tests and evidence
5 Derek Song (24833978)	F5 Buyer journey workspace	F5 workflow, endpoint tests and evidence

2 Final repository structure

Path	Contents	Release
`student-1/` … `student-5/`	Feature slices: UI, API, database service, tools, corpus	0–1
`student-1/config/multi-agent/`, `student-1/tests/endpoints/`	F1 workflow template and endpoint tests	2
`ai-services/ai-mode/`, `agent-core/`	AI-mode run API and the shared agentic loop	0
`ai-services/mcp-server/`, `rag-server/`	Tool transport and local retrieval	1
`ai-services/multi-agent-server/`	Planner, Worker, Reviewer, decisions, JSONL audit, CLI	2
`scripts/devtools/review/`	Review modes and release decision	2
`shared/frontend/multi-agent/`, `shared/testkit/`	Shared workflow panel; endpoint client, JUnit summary, `FakeMultiAgentServer`	2
`.pre-commit-config.yaml`, `scripts/security/`	Security hooks and report	2
`.github/workflows/`	`student-N.yml`, `integration-ci.yml`; `cloud-deployment.yml`	0–2
`deployment/azure/`, `docker-compose.azure.yml`	Bicep, VM scripts, TLS edge, systemd AI units	2
`docs/release-2/evidence/`	Security, CI, multi-agent, review, local and cloud evidence	2

3 Release 2 design flow

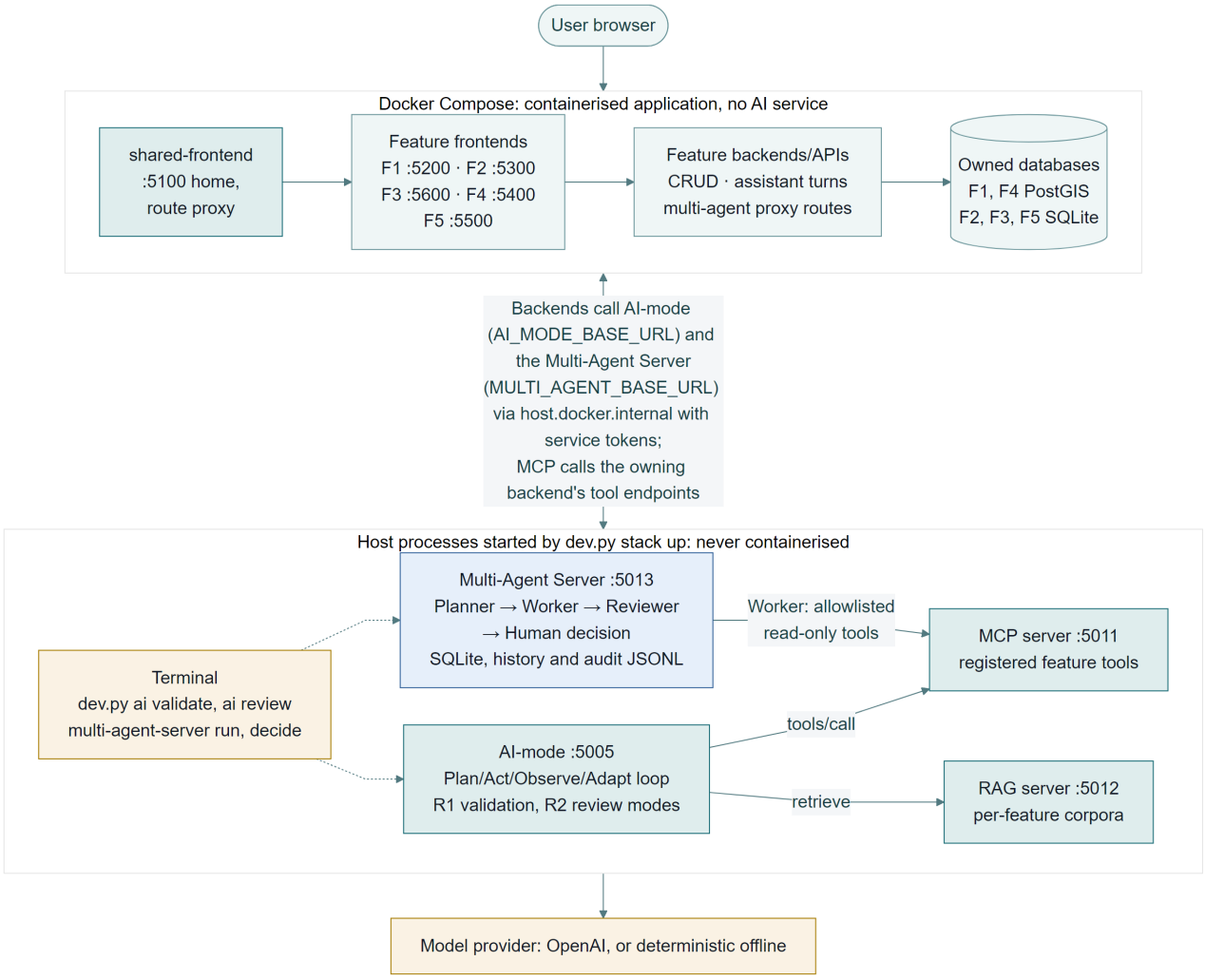


Figure 1 Release 2 integrated architecture

CRUD requests pass from the browser through the shared edge to a feature frontend, its backend/API and its own database service. Assistant turns keep the Release 1 path through AI-mode to the feature's MCP tools and RAG corpus. A backend with a workflow also proxies the Multi-Agent Server over HTTP with a service token, and the shared panel shows each stage up to the human decision. No AI service opens a feature database or runs in Docker.

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template
1	:5200 → `/api/data-platform/v1` → `f1-postgres`	14	19	`f1-release-readiness-review` (2 read-only tools)
2	:5300 → `/api/market-intelligence/v1` → `f2-db-api`	3	3	Not registered yet
3	:5600 → `/api/suburb-analytics/v1` → `f3-database`	4	8	Not registered yet

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template
4	:5400 → `/api/due-diligence/v1` → `f4-postgres`	3	3	Not registered yet
5	:5500 → `/api/buyer-workspaces/v1` → `f5-db-api`	5	5	Not registered yet

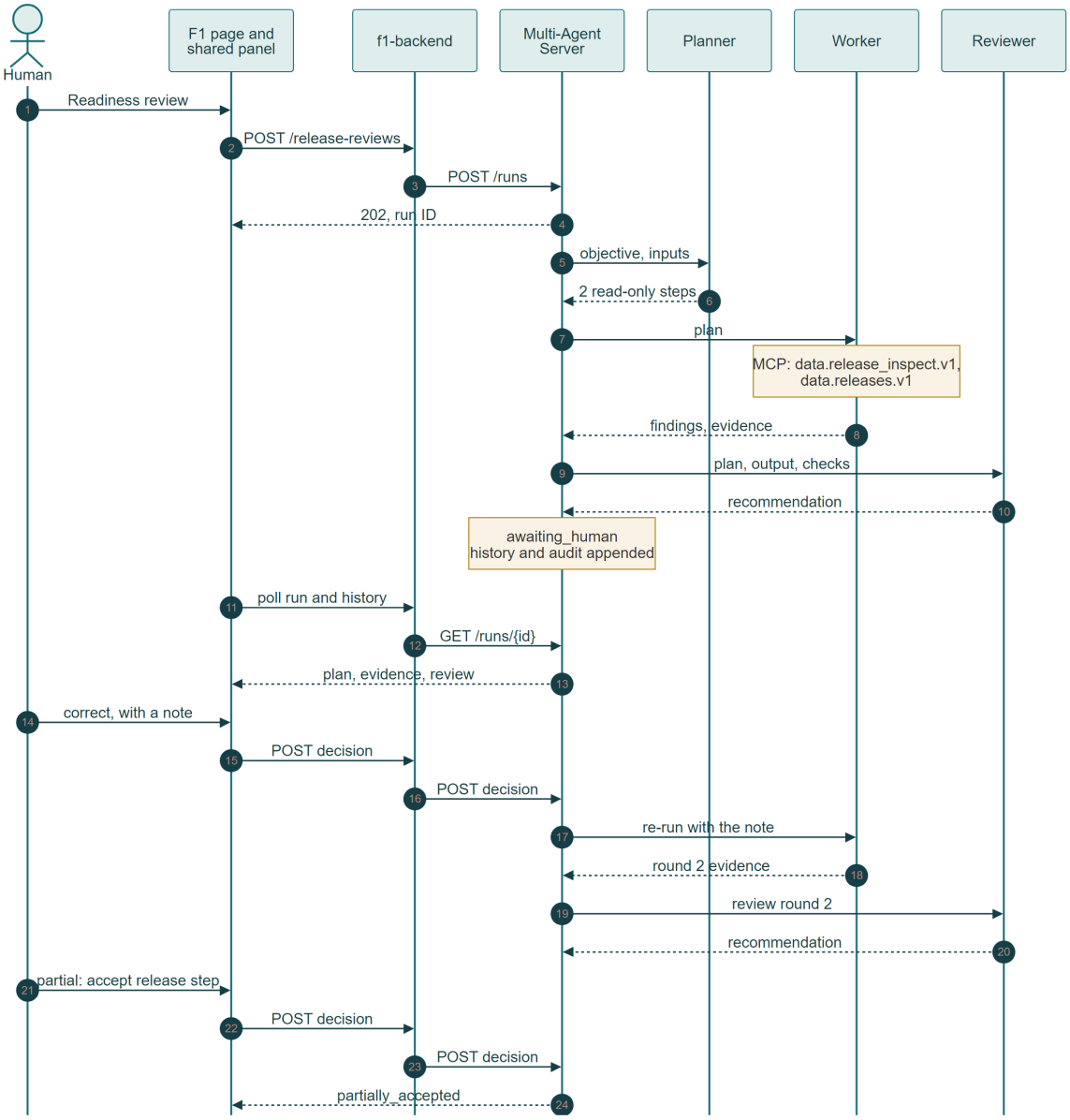


Figure 2 Multi-agent flow, using the Feature 1 readiness review

Student 1

`f1-release-readiness-review` recommends whether a candidate dataset release is ready to publish. Seven proxy routes under `/api/data-platform/v1/release-reviews` fix the template server-side and hide other features' runs. The allowlist omits `data.release_publish.v1`, so a review never publishes.

Student 2

TO COMPLETE / BURHAN NAEEM

Describe F2's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 3

TO COMPLETE / JAMES HUANG

Describe F3's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 4

TO COMPLETE / MICHAEL WHITE

Describe F4's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 5

TO COMPLETE / DEREK SONG

Describe F5's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

4 Cloud deployment and CI/CD architecture

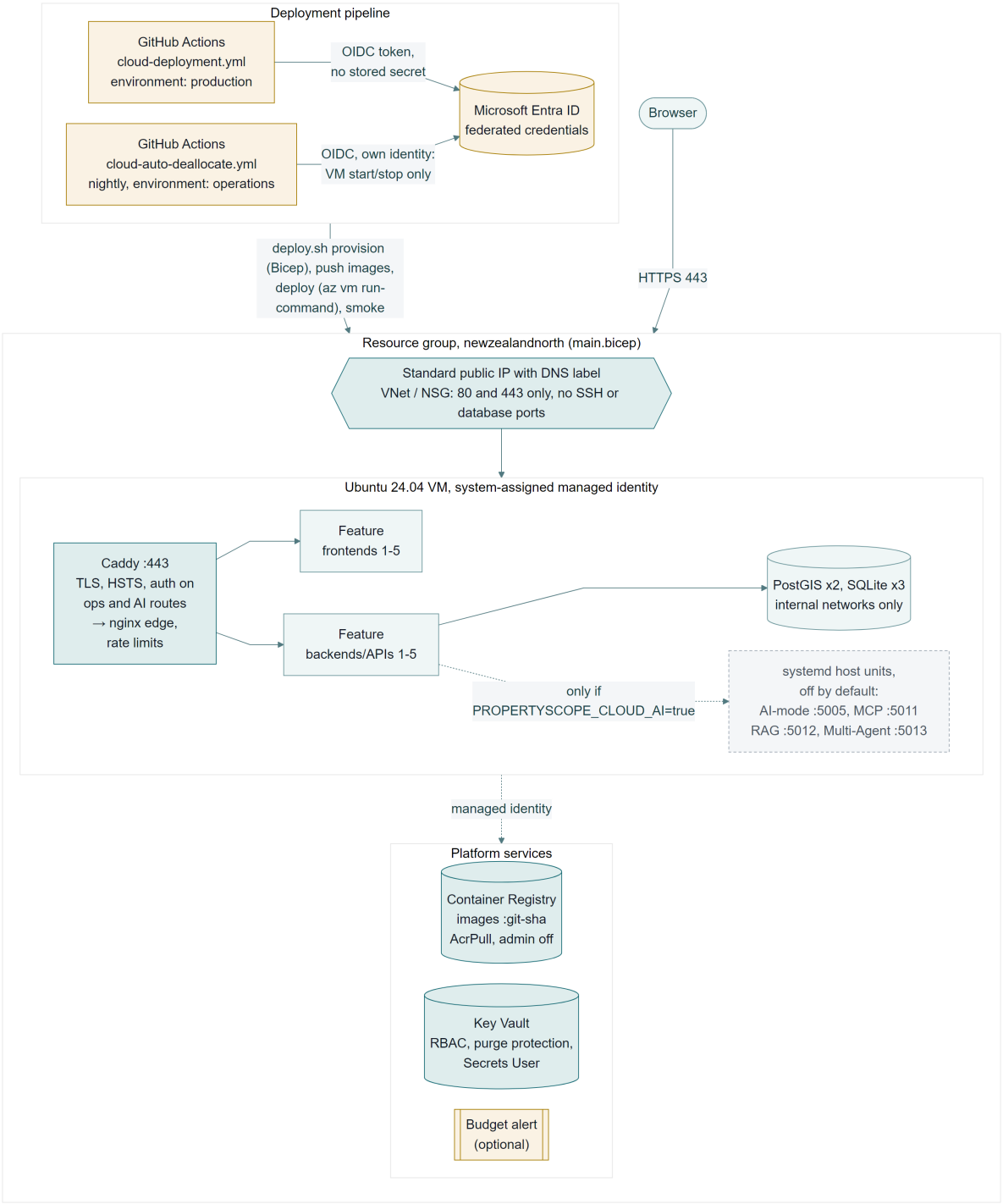


Figure 3 Azure reference architecture

We chose Azure (ADR-048): one Ubuntu VM (`Standard_D4as_v4`, `newzealandnorth`) runs the Compose model with a production override, because the AI tier must stay a host process. Its managed identity reads Container Registry and Key Vault; GitHub signs in with OIDC. The NSG admits only 80 and 443, Caddy terminates TLS, and the four AI systemd units stay off unless `PROPERTYSCOPE_CLOUD_AI=true`.

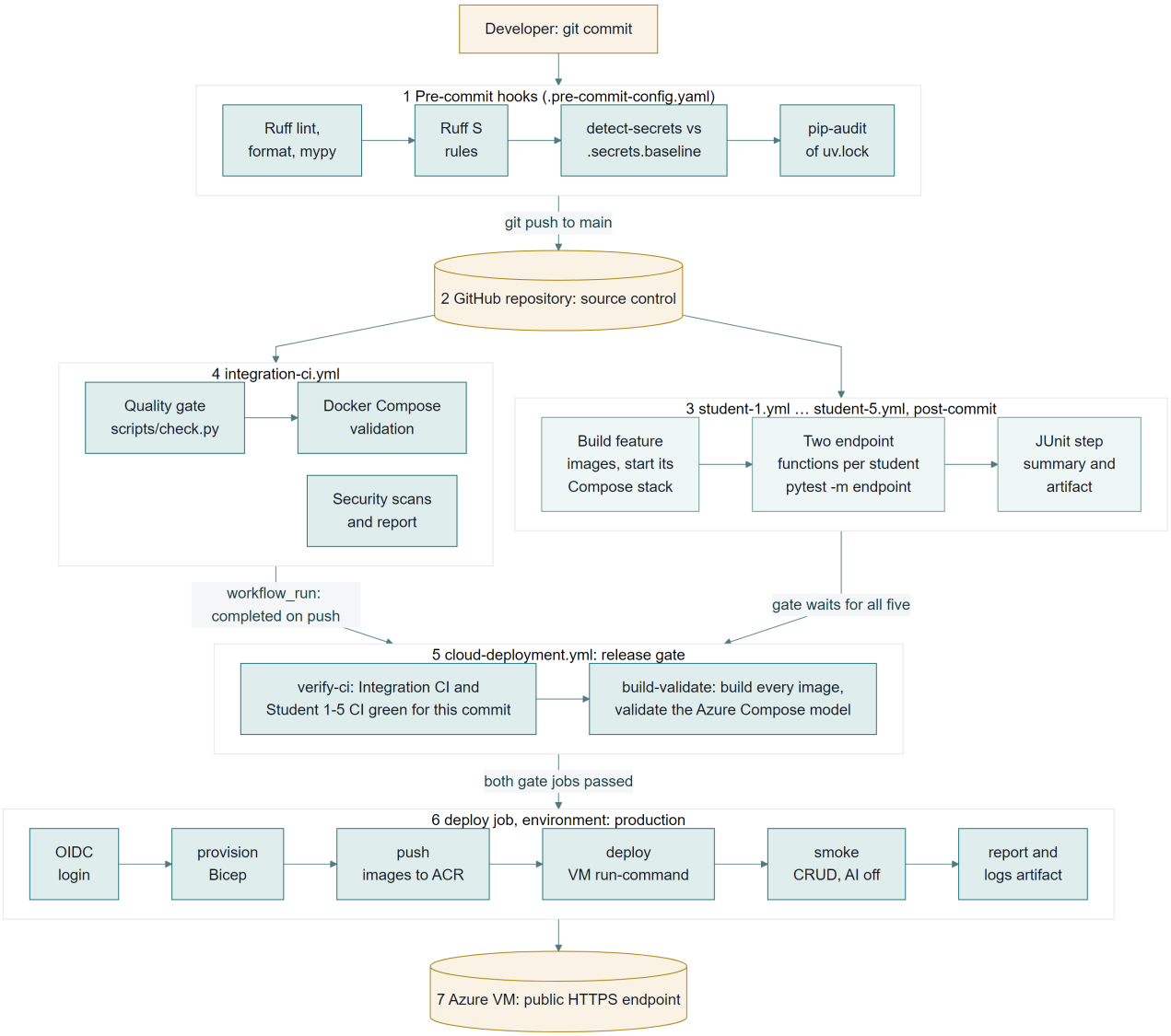


Figure 4 DevOps flow from commit to Azure

A commit runs the pre-commit hooks; a push to `main` runs `integration-ci.yml` and the five `student-N.yml` workflows, whose endpoint tests are each owner's deliverable. `cloud-deployment.yml` gates the release: `verify-ci` waits until all six pass for the commit and `build-validate` builds and validates every image. Only then does `deploy` run these reusable scripts in the protected `production` environment:

Script or template	Role
`deploy.sh provision`, `secrets`	Resource group, `main.bicep`, missing Key Vault secrets
`deploy.sh push`, `deploy`	Images to ACR; VM converged through `az vm run-command`
`deploy.sh smoke`, `status`, `logs`	Public smoke test and VM state
`deploy.sh ai on/off`, `ai-probe`	Bonus AI tier and its B1–B4 probe
`deploy.sh validate-endpoint`, `validate-data`	B5 and B6 checks
`main.bicep`, `modules/`	Network, registry, Key Vault, VM, RBAC, shutdown, budget
`validate-iac.sh`	Offline Bicep, shell, Caddy and nginx checks (CI `iac-lint`)

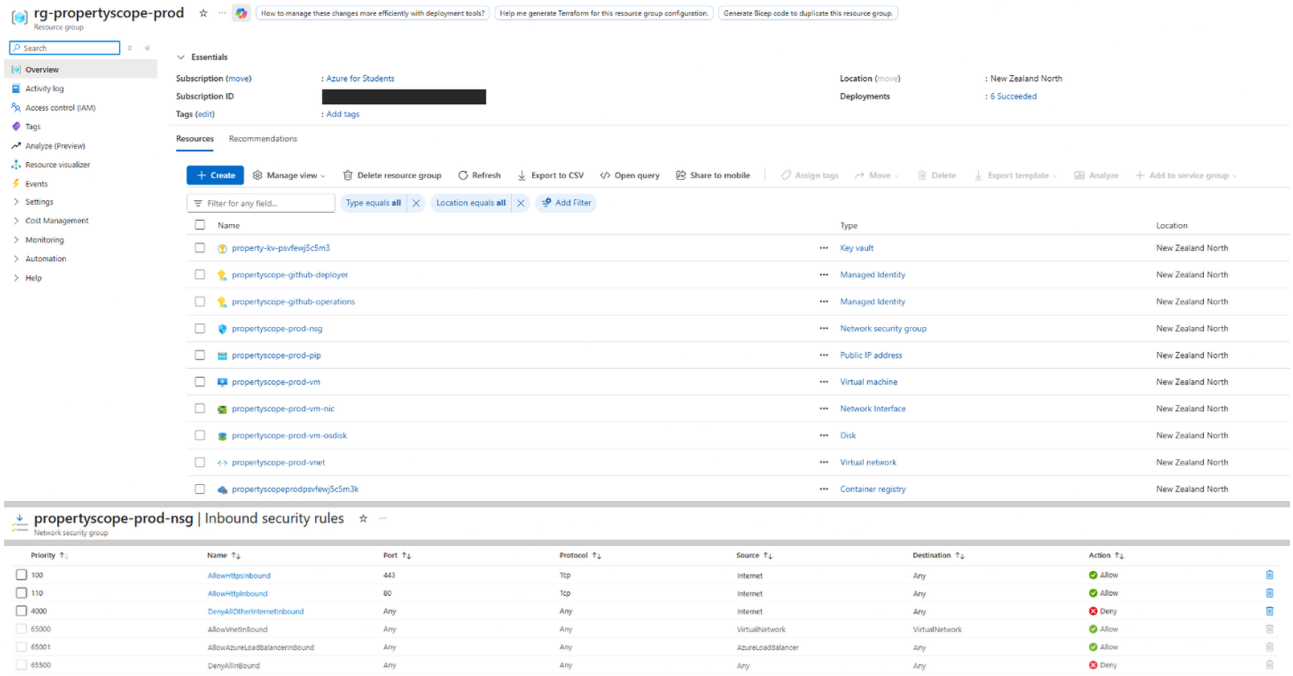


Figure 5 Deployed resources and inbound rules

5 Pre-commit security testing

`.pre-commit-config.yaml` runs three security hooks on every `git commit`: Ruff S (flake8-bandit) rules, detect-secrets against the reviewed `.secrets.baseline`, and pip-audit of `uv.lock`. `integration-ci.yml` repeats them on every push. `dev.py security report` records results per slice and owner:

Security report: overall **PASS**, commit `1a68f0a`, 2026-10-16.

Scan	Result	Findings
Ruff	**PASS**	0 open; 27 fixed; 132 justified with `# noqa`; 85 pending owner
detect-secrets	**PASS**	0 new; 0 confirmed; 0 unaudited; 56 reviewed false positives in `.secrets.baseline`
pip-audit	**PASS**	126 locked packages audited; 0 open; 19 accepted risk(s) in 2 package(s)

Slice	Owner	Ruff open	Fixed	Noqa	Pending owner	Secrets to act on
Shared platform	Matthew Shelton	0	18	111	0	0
Student 1	Matthew Shelton	0	9	21	0	0
Student 2	Burhan Naeem	0	0	0	8	0
Student 3	James Huang	0	0	0	10	0
Student 4	Michael White	0	0	0	8	0
Student 5	Derek Song	0	0	0	59	0

In the recorded commit run, detect-secrets first blocked the commit; after seven false positives were reviewed into the baseline, it passed:

```
security: Ruff flake8-bandit rules (S).....................Passed
security: detect-secrets against .secrets.baseline.............Passed
security: pip-audit of locked dependencies (uv.lock)............Passed
```

TO COMPLETE / MATTHEW SHELTON

Regenerate the security report on main at the evidence freeze; the committed one scanned a release-2/wrap-up-shared-f1 branch commit

TO COMPLETE / BURHAN NAEEM

Fix or justify F2's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / JAMES HUANG

Fix or justify F3's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / MICHAEL WHITE

Fix or justify F4's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / DEREK SONG

Fix or justify F5's pending-owner Ruff S findings, then regenerate the report

6 Post-commit CI/CD testing

Each owner's `student-N.yml` should start the feature's Compose stack with AI disabled, run `pytest -m endpoint` on two endpoint functions and publish the Shared test kit's JUnit summary (at least two passes). `collect_ci_evidence.py` keeps each run's record, reports and log.

Student	`student-N.yml` run	Conclusion	Commit	Endpoint tests
1	#387	success	`817ef77`	46 passed, 0 failed (2 reports) (record)
2	Pending	-	-	-
3	Pending	-	-	-
4	Pending	-	-	-
5	Pending	-	-	-

TO COMPLETE / BURHAN NAEEM

Record a green `student-2.yml` run on main with `collect_ci_evidence.py --student 2`

TO COMPLETE / JAMES HUANG

Record a green `student-3.yml` run on main with `collect_ci_evidence.py --student 3`

TO COMPLETE / MICHAEL WHITE

Record a green `student-4.yml` run on main with `collect_ci_evidence.py --student 4`

> **TO COMPLETE / DEREK SONG**
>
> Record a green `student-5.yml` run on main with `collect_ci_evidence.py --student 5`

Student 1

`student-1.yml` loads fixture data, then tests the feature origin (:5200) and the shared edge (:5100):

1. `GET /api/data-platform/v1/properties/search`: contract-valid ranked matches for an accepted address; invalid queries return structured problems.
2. `/api/data-platform/v1/sources`: create, list, read, update and delete a source definition; invalid, duplicate and non-JSON bodies are rejected and not persisted.

Suburb-map tests also run; two skip without a suburb-boundaries release.

Student 2

> **TO COMPLETE / BURHAN NAEEM**
>
> Name F2's two endpoint functions and what each test checks (about 60 words)

Student 3

> **TO COMPLETE / JAMES HUANG**
>
> Name F3's two endpoint functions and what each test checks (about 60 words)

Student 4

> **TO COMPLETE / MICHAEL WHITE**
>
> Name F4's two endpoint functions and what each test checks (about 60 words)

Student 5

> **TO COMPLETE / DEREK SONG**
>
> Name F5's two endpoint functions and what each test checks (about 60 words)

7 Testing review

`dev.py ai review testing` checks the security report and each `ci/student-N.md` against a deterministic checklist and reviews them through the shared loop. The stricter verdict wins, and each review appends to the validation log.

Verdict	Checklist	Engine	Loop run	Date	Validation log
fail	10/32 passed	AI-mode, openai / gpt-5.6-terra	`0b46ed6a`	2026-10-16	3 entries

Latest validation-log entry: `review_completed`, review `71c1b5d1`, `review-testing.v1`, bundle `62378459`, 10 passed / 22 failed, verdict **fail**, 2026-10-16.

> All three pre-commit security scans were recorded. Student 1 supplied a successful CI run with endpoint coverage and no JUnit failures or errors; students 2–5 supplied no CI evidence. (full review)

TO COMPLETE / MATTHEW SHELTON

Re-run the testing review once all five CI records exist; the committed review predates them

8 Local deployment evidence

`dev.py stack up` starts the four host AI services, then the Compose application; backends reach them through `host.docker.internal`. `capture_local_deployment.py` checks the running system, including that each AI service is a host process, not a container. The committed capture recorded the host services through managed log files; the freeze recapture runs each in its own foreground terminal.

Local deployment report: **PASSED**, 109 of 109 checks passed, captured 2026-10-16 at commit `1a68f0a`.

Check group	Passed	Failed
Compose containers	22	0
Host AI services	9	0
Edge, frontends and health	13	0
CRUD round trips	20	0
Data	1	0
Direct AI probes	32	0
Multi-Agent terminal run	6	0
Evidence files	6	0

Terminal	Ran as	Command	Log
Docker Compose stack	foreground	`uv run scripts/dev.py stack up; stack logs`	`terminal-1-compose.log`
AI-mode host process	managed	`uv run scripts/dev.py ai serve ai-mode`	`terminal-2-ai-mode.log`
MCP host process	managed	`uv run scripts/dev.py ai serve mcp`	`terminal-3-mcp.log`

Terminal	Ran as	Command	Log
RAG host process	managed	`uv run scripts/dev.py ai serve rag`	terminal-4-rag.log
Multi-Agent host process	managed	`uv run scripts/dev.py ai serve multi-agent`	terminal-5-multi-agent.log
System test	foreground	`uv run python scripts/capture_local_deployment.py`	terminal-6-system-test.log

TO COMPLETE / MATTHEW SHELTON

Recapture the local deployment report on main at the evidence freeze; this capture ran on a branch commit

9 Cloud deployment report and review

`deploy.sh smoke` checks the home page, every feature route, create, read and delete per feature, and that AI-mode answers `503 ai_disabled`; all 37 checks passed. `dev.py ai review cloud` reviews the evidence, then a named person records the release decision with `ai review decide`, which refuses if the evidence changed.

Item	Recorded
Deployment	**SUCCEEDED** (report)
Public URL	https://propertyscope-nsw-g20.newzealandnorth.cloudapp.azure.com
Commit / image tag	`3c25c083e09e5f170767e63655593c0e5b9f1e27` / `3c25c083e09e5f170767e63655593c0e5b9f1e27`
Workflow run	https://github.com/MattShelton04/41026ASDProject/actions/runs/39112162750
AI tier	flag off (PROPERTYSCOPE_CLOUD_AI), edge reports disabled
Human release decision	**RELEASE** by Matthew Shelton, 2026-10-19 (record)

Run 39112162750 job timings: `verify-ci` 15 s, `build-validate` 42 s, `deploy` 10 min 39 s (run record, step logs, `vm-deploy.log`). Run 39098352903 also ran the Bicep provision step (1 min 29 s).

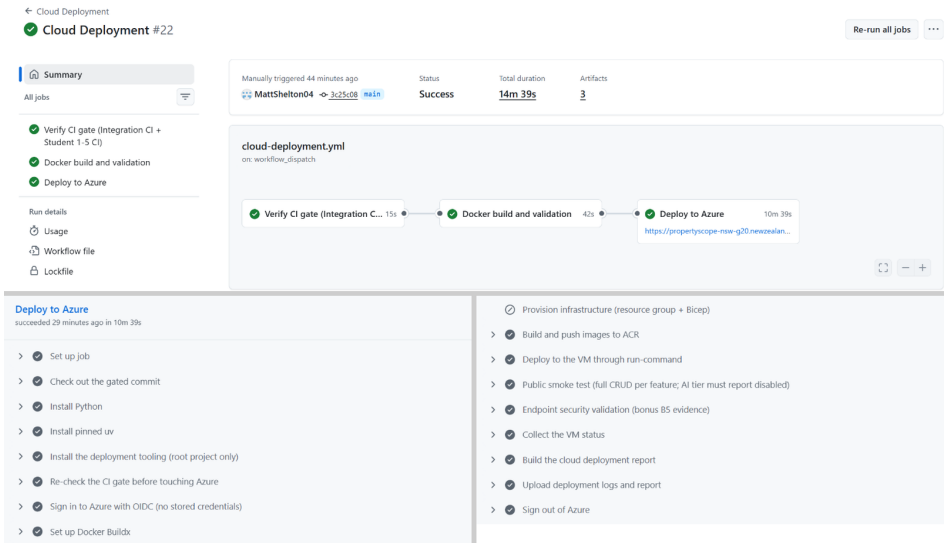


Figure 6 Gated cloud deployment run

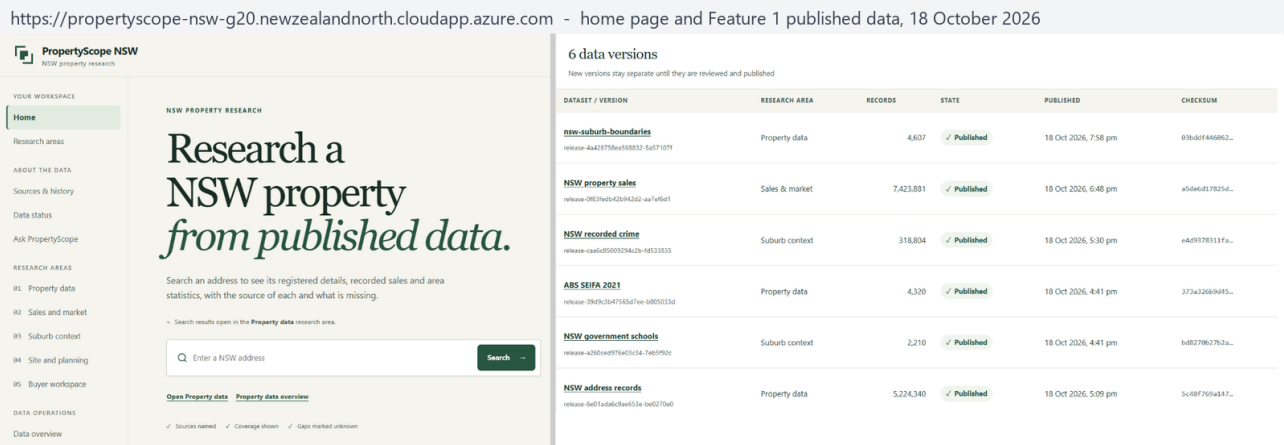


Figure 7 Deployed application with real Feature 1 data

Verdict	Checklist	Engine	Loop run	Date	Validation log
pass	-	deterministic, deterministic-review / checklist-review.v1	`22a7614b`	2026-10-18	3 entries

Latest validation-log entry: `human_release_decision` (**release**), review `44929d5d`, `review-cloud.v2`, bundle `b0362e89`, 18 passed / 0 failed, verdict **pass**, 2026-10-19.

> Cloud Deployment Report Review: 18 of 18 checks passed, 0 failed (0 required), across 10 evidence file(s), 0 of them missing or unreadable. (full review)

10 Multi-agent workflow evidence

The server keeps runs in SQLite, state transitions in `workflow_history.jsonl`, and handoffs, model and tool calls and decisions, with actor and time, in `coordination_audit.jsonl`; `multi-agent-server export` copies a run into the evidence.

Student	Run	History events	Audit events	Handoffs	Tool calls	Decisions	Final state
1	`18740ca5`	8	24	7	4	correct → partial	`partially_accepted`
1	`c17c66cc`	8	25	7	4	correct → partial	`partially_accepted`
2	Pending	-	-	-	-	-	-
3	Pending	-	-	-	-	-	-
4	Pending	-	-	-	-	-	-
5	Pending	-	-	-	-	-	-

TO COMPLETE / BURHAN NAEEM

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-2/<run> --server`

TO COMPLETE / JAMES HUANG

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-3/<run> --server`

TO COMPLETE / MICHAEL WHITE

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-4/<run> --server`

TO COMPLETE / DEREK SONG

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-5/<run> --server`

Multi-agent workflow review

`dev.py ai review multi-agent` checks every export for all four stages, legal transitions, attributed human decisions and allowlisted tool calls.

Verdict	Checklist	Engine	Loop run	Date	Validation log
fail	7/35 passed	AI-mode, openai / gpt-5.6-terra	`bf3f4052`	2026-10-16	6 entries

Latest validation-log entry: `review_completed`, review `af46b4f5`, `review-multi-agent.v1`, bundle `a0453adf`, 7 passed / 28 failed, verdict **fail**, 2026-10-16.

> Student 1 has complete, correlated evidence for the required multi-agent workflow controls. (full review)

Student 1

UI run `18740ca5` (10 October, model provider) reviewed a real 680-record ABS CPI candidate. Round 1: the Reviewer recommended approve; the human chose correct, asking for completeness against ABS. Round 2: the Reviewer recommended correct, as read-only tools cannot fetch ABS totals; the human accepted step `release` only, publishing nothing. Terminal run `c17c66cc` used the same template.

```
$ multi-agent-server export 18740ca5-5fec-482a-a279-fe179a6d25a6
01:18:30 r1              - -> planning           multi-agent-server
01:18:35 r1       planning -> working            planner
01:18:39 r1        working -> reviewing          worker
01:18:44 r1      reviewing -> awaiting_human     reviewer
01:20:46 r2 awaiting_human -> working            matthew
01:20:51 r2        working -> reviewing          worker
01:20:57 r2      reviewing -> awaiting_human     reviewer
01:21:32 r2 awaiting_human -> partially_accepted matthew
```

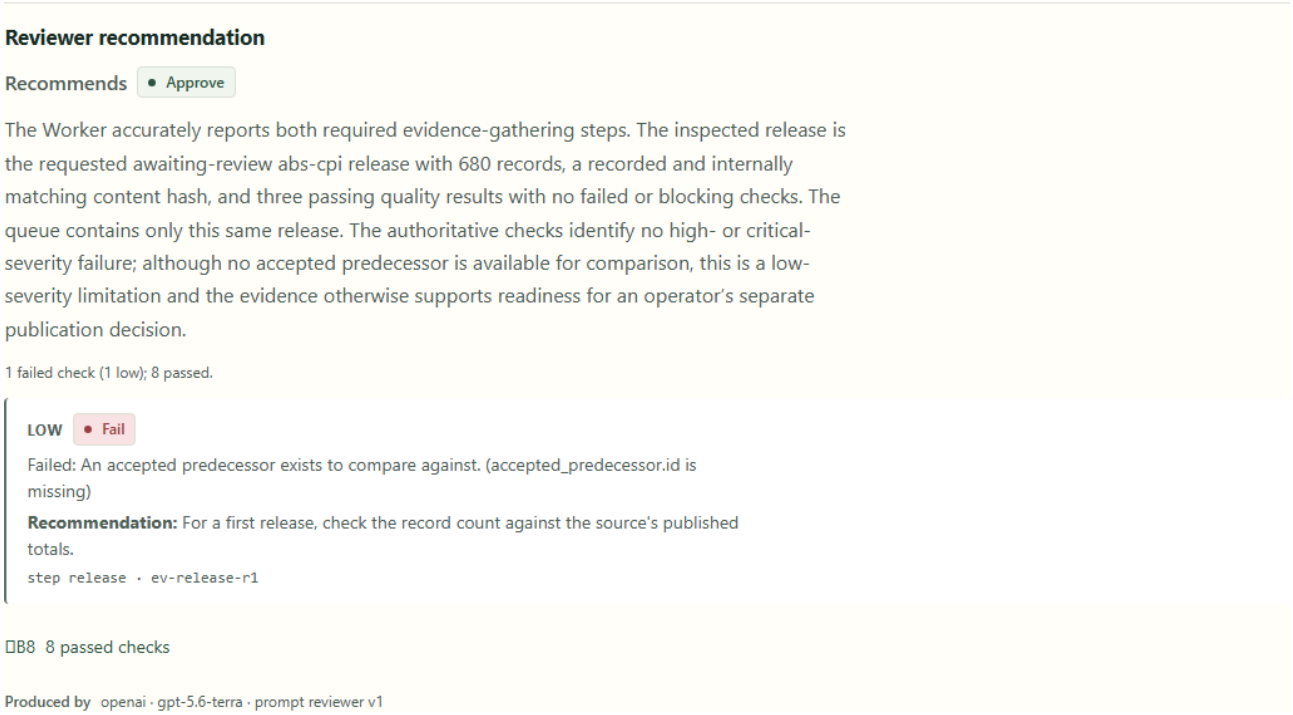


Figure 8 F1 Reviewer recommendation, round 1

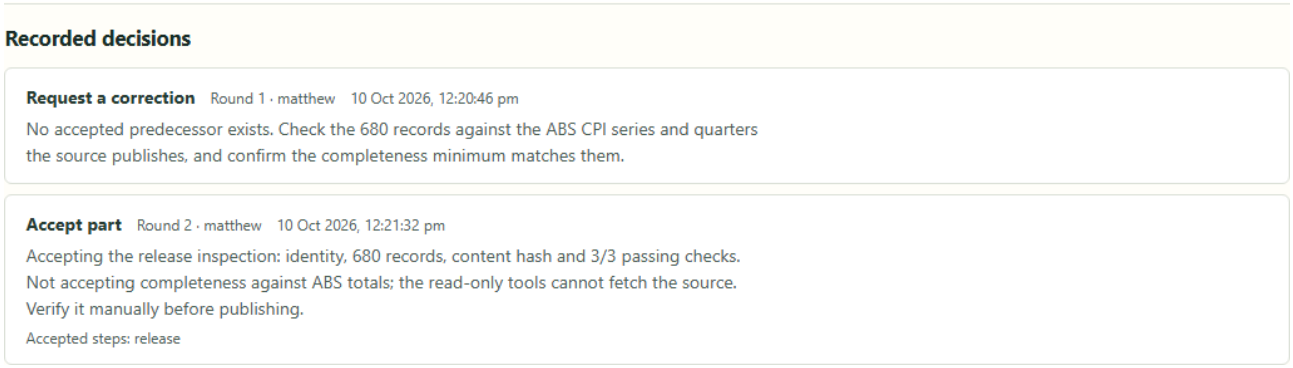


Figure 9 F1 recorded human decisions

Student 2

TO COMPLETE / BURHAN NAEEM

Summarise F2's exported run in about 50 words: input, recommendation and human decision

Student 3

TO COMPLETE / JAMES HUANG

Summarise F3's exported run in about 50 words: input, recommendation and human decision

Student 4

TO COMPLETE / MICHAEL WHITE

Summarise F4's exported run in about 50 words: input, recommendation and human decision

Student 5

TO COMPLETE / DEREK SONG

Summarise F5's exported run in about 50 words: input, recommendation and human decision

11 GitHub commit logs

Commit history: https://github.com/MattShelton04/41026ASDProject/commits/main. Matthew's commits: https://github.com/MattShelton04/41026ASDProject/commits/main?author=MattShelton04; each student's GitHub username filters the same way, and Section 12 lists their commits.

12 Contribution logs

Student 1 Matthew Shelton

Date	Commit	Release 2 work
8–10 Oct	36a4c26, f86f787, bef57da, 06e0ed0	Multi-Agent Server, review modes, security scans, test kit, panel, F1 review
15–17 Oct	d449a43, 07b1c81, 72f3272, fdc5111, 93d0f53	CI and local evidence, report builder, release gate
17 Oct	5e90231, ad2b443, 71764a5, cd8e56d	Azure switch-on, site login, security fixes
18 Oct	b638c86, 53a2143, 65de8f6, b71c49c, 3c25c08	F1 suburb-map API, real data on Azure
18 Oct	c1dab31, 3ea2932, 817ef77, 0cf5e3f	Cloud AI Tier, cloud and bonus evidence, release decision

Student 2 Burhan Naeem

TO COMPLETE / BURHAN NAEEM

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 3 James Huang

TO COMPLETE / JAMES HUANG

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 4 Michael White

> **TO COMPLETE / MICHAEL WHITE**
>
> Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 5 Derek Song

> **TO COMPLETE / DEREK SONG**
>
> Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

13 Bonus evidence

We claim B1–B6. B1–B4 ran through the public edge in Cloud AI Tier run 39119158736 (`817ef779`, 18 October).

Tier	Result and evidence	Run or source
B1 AI-mode	PASS	AI-mode run `387591f0`
B2 MCP	PASS	`data.releases.v1` over MCP, same run
B3 RAG	PASS	Run `56c85266`, 2 citations
B4 Multi-Agent	PASS	Planner → Worker → Reviewer run `2058658b`, `awaiting_human`
B5 Endpoint security	52 PASS, 0 FAIL	Cloud Deployment run 39112162750
B6 Data security	43 PASS, 0 FAIL	Laptop, 18 October

The showcase video also demonstrates each claimed bonus through the Azure UI.

14 Showcase video

> **TO COMPLETE / GROUP**
>
> Link the published showcase video (at most 10 minutes) and record which students participated